Barriers to Mobile Banking and Digital Wallet Adoption Among University and College Students in Pakistan

Okasha Saeed
Department of Financial Technology
FAST NUCES Islamabad
Islamabad, Pakistan
i235556@isb.nu.edu.pk

***Abstract*—This paper presents the evaluation of a machine-learning classification system designed to identify the main barriers stopping students in Pakistan from adopting mobile banking and digital wallet services. Using a simulated survey dataset of 200 student responses, two classification models namely Logistic Regression (LR) and Support Vector Machine (SVM) with an RBF kernel—were trained and used. Barriers were categorised into four classes: Technological, Psychological, Financial, and Social. Results show that Logistic Regression achieved the highest accuracy of 92.5 %, outperforming SVM (85.0 %) and the baseline (25.0 %). The results show that machine learning approaches can classify student barriers and support targeted interventions.**

***Index Terms*—mobile banking, digital wallet adoption, Pakistan, machine learning, classification, barriers, financial inclusion, TAM, UTAUT**

I. Introduction

Mobile banking applications and digital wallets have transformed financial services globally, and Pakistan is no exception to this trend. Over the last five years, applications such as JazzCash, Easypaisa, and SadaPay have expanded widely across the country, offering users access to financial services without the need of a traditional bank account. Despite this growth, a large segment which is the university and college going students have hesitated in adopting these new technologies. [1].

The widespread availability of mobile banking apps has not translated into similar usage among the student population. A significant percentage of students still rely on cash for everyday transactions even when digital alternatives are easily accessible. The underlying reasons for this behaviour remain unclear. Factors such as ease of use, security concerns including fear of hacking or fraud, unstable internet access, limited financial literacy, and social norms may all influence a student's decision to adopt or avoid mobile banking applications [3].

This problem carries significance in a country where a large and growing population of young, technically literate students has access to smartphones/computers yet remains excluded from the digital economy. Students represent a large demographic for financial inclusion efforts because their early adoption of mobile banking habits can have lasting effects on economic activity [2]. Research has consistently shown that once young users form digital payment habits, those habits tend to persist into working life and drive wider adoption across society [14].

Existing research on mobile banking adoption has largely focused on general adult populations or working professionals. There is a gap in the research targeting university and college students in Pakistan as a specific group with unique patterns, technological literacy levels, trust levels, and social influences [11]. Several studies have applied the Technology Acceptance Model (TAM) and the Unified Theory of Acceptance and Use of Technology (UTAUT) to explain adoption behaviour in other countries, but these frameworks have rarely been applied in a student specific Pakistani context [6], [9].

Understanding the barriers specific to this group is essential for designing effective financial inclusion strategies. Fintech companies, educational institutions, and policy makers will all benefit from clearer reasons as to what prevents students from making the transition from cash to digital payments [4], [5].

The objective of this research paper is therefore to identify and analyse the barriers that prevent Pakistani university students from adopting mobile banking and digital wallet applications. A survey based methodology has been used , and machine learning classification models are applied to categorise students by their dominant barrier type. The findings are intended to inform targeted recommendations for fintech providers, educational institutions, and policymakers.

The rest of this paper is organised as follows. Section II presents the problem statement. Section III identifies the research gap. Section IV reviews related literature. Section V describes the proposed methodology. Sections VI and VII present the system architecture and workflow diagrams. Section VIII covers implementation details. Section IX describes the dataset. Section X outlines the experiment setup. Section XI defines evaluation metrics. Section XII presents results. Section XIII showcases results. Section XIV talks about the limitations. Section XV concludes the paper and Section XVI outlines future work.

II. Problem Statement

Mobile banking applications and digital wallets have transformed the world globally, and specifically Pakistan. In the last 5 years, apps such as JazzCash, Easypaisa, and SadaPay have expanded widely, offering students convenient access to financial services without the need for a regular bank account. Despite this, a large segment of university and college-going students in Pakistan are hesitating to use these tools [1].

Even though mobile banking apps are available, not all students use them regularly. A large number of students prefer using cash instead of mobile apps. The reasons behind this are not clearly understood. Factors such as ease of use, security issues (hacking, etc.), and internet access may influence the decision of students to adopt banking applications [3].

This problem is particularly significant in a country where a large population of students has a strong need for these tools and is technically capable of using these applications. Students represent an important demographic for financial inclusion, as their early adoption of microfinance banking habits can have a long-term effect on economic activity [2].

III. Research Gap

While a substantial body of research exists on mobile banking adoption, the majority of these studies focus on identifying factors that increase adoption using frameworks such as TAM [9] and UTAUT [6]. These models have been applied in countries such as India, China, Malaysia, and several Western nations, but are not widely applied or examined in the Pakistani context [2].

Moreover, existing research places focus on the general population or banking customers rather than specifically targeting university students as a distinct group. Students have unique behavioral patterns, technology literacy levels, trust dispositions, and social norms that differ considerably from working adults [11].

There is limited research examining the full spectrum of barriers students face technological, psychological, financial, and social [5]. This gap demonstrates the need for targeted research aimed at identifying the specific barriers preventing Pakistani students from fully adopting mobile banking applications.

IV. Literature Review

The literature on mobile banking and digital wallet adoption spans multiple geographic and demographic contexts. Early foundational work by Luarn and Lin [9] established that trust is one of the most important determinants of mobile banking adoption intention. Their study demonstrated that unless users believe a system is reliable and secure, they are unlikely to adopt it regardless of its usefulness. This finding has been replicated across later studies and contexts.

Ahmed et al. [2] conducted a specific case study of Pakistan and identified trust, perceived usefulness, and ease of use as the three primary factors driving IT-based banking adoption. Their work is relevant to the current study because it establishes a baseline for understanding user attitudes toward digital financial tools. Similarly, Iqbal et al. [1] examined JazzCash and Easypaisa specifically and found that usability and trust remain the most significant predictors of adoption in the Pakistani market.

Research applying UTAUT has produced insights into social dimensions of adoption. Santoso et al. [6] found that social influence and facilitating conditions were the strongest predictors of ewallet usage . Al-Qudah et al. [7] extended this line of inquiry to a pandemic context in Jordan, finding that perceived usefulness and security concerns were amplified during COVID-19, leading to accelerated adoption. Octaviani and Rustiana [8] applied UTAUT2 in Indonesia and identified motivation and habit as additional significant reasons, suggesting that enjoyment of the technology itself plays a role beyond the functional concerns.

Studies examining financial inclusion have highlighted the importance of designing mobile banking services for underserved populations. Mas and Radcliffe [12] analyzed how mobile banking features could be adapted to serve low income communities, providing designs that are applicable to student populations with limited financial resources. Raza et al. [5] proposed a framework for digital wallet adoption specifically focused on financial inclusion in developing countries, identifying infrastructure limitations and literacy as the main barriers.

Studies reveal the diversity of adoption challenges. Alotaibi and Roussinov [13] reviewed the mobile payment landscape in Saudi Arabia and identified infrastructure gaps and low public trust as the primary barriers. Hussain et al. [15] examined digital wallet adoption in Bahrain using CFA and SEM and found that security, convenience, and social norms were the main drivers. Lim et al. [14] surveyed millennials in Malaysia and found that perceived usefulness, ease of use, and social influence significantly predicted adoption intent, results directly comparable to a student population in Pakistan.

Table I presents a focused comparison of the most directly relevant studies, highlighting their methodology, context, and key findings.

Table II provides a broader summary of all 20 papers reviewed for this study.

Collectively, the literature confirms that mobile banking adoption is a multi dimensional event effected by technological, psychological, financial, and social factors. However, no existing study has applied a machine learning approach to categorise Pakistani students by their dominant barrier type. This study does exactly that.

V. Proposed Methodology

This study employs a quantitative survey based research methodology to analyze the barriers preventing university and college students in Pakistan from adopting mobile banking applications and digital wallets. The methodology is divided into five phases.

A. Data Collection

Primary data will be collected through a survey distributed among university and college students in Islamabad. The survey

TABLE I
LITERATURE COMPARISON TABLE

#	Study	Context	Method	Framework	Key Finding
1	Iqbal et al. [1] (2022)	Pakistan	Survey	TAM	Usability and trust are the top adoption factors for JazzCash and Easypaisa.
2	Ahmed et al. [2] (2011)	Pakistan	Case Study	TAM	Trust, usefulness, and ease of use drive IT banking adoption in Pakistan.
3	Santoso et al. [6] (2024)	Indonesia	Survey	UTAUT	Social influence and facilitating conditions are the strongest predictors.
4	Lim et al. [14] (2022)	Malaysia	Survey	TAM/UTAUT	Ease of use and social influence predict millennial wallet adoption.
5	Hussain et al. [15] (2021)	Bahrain	SEM/CFA	UTAUT2	Security, convenience, and social norms drive digital wallet adoption.
6	Raza et al. [5] (2023)	Developing nations	Conceptual	Custom	Infrastructure and literacy are central barriers to financial inclusion.
7	Luarn & Lin [9] (2009)	Taiwan	Survey	TAM	Trust is the most critical determinant of mobile banking adoption.

is designed based on established frameworks including TAM and UTAUT. The questionnaire covers ease of use, security, internet access, social influence, trust, and financial literacy. A minimum of 200 responses will be recorded via Google Forms distributed through university WhatsApp groups.

B. Data Cleaning

Once raw data is collected, it will be imported into Python for cleaning and preparation. Incomplete or duplicate answers will be removed, and missing values will be handled. Any response with more than 20 % missing data will be discarded. Responses on a 1–5 scale will be normalized using min-max scaling. Outliers will be detected using z-scores and removed.

C. Barrier Identification

Key barriers will be identified from the survey data using analysis to group related variables into four categories: Technological, Psychological, Financial, and Social. The output of this phase is a labelled dataset for model training.

D. Model Design and Classification Framework

A machine-learning classification framework will categorise students into groups based on their primary barrier. Logistic Regression and SVM will be trained and tested on an 80/20 split (160 training, 40 testing). The best model will be selected based on F1-Score.

E. Result Interpretation and Decision Output

Results will be visualised using bar charts and pie charts, compared with the reviewed literature, and translated into actionable recommendations for microfinance banks and educational institutions.

VI. SYSTEM ARCHITECTURE DIAGRAM

The system architecture is illustrated in Fig. 1. It consists of six layers: (1) Data Source—survey responses from university students; (2) Data Input—CSV import into Python from Google Forms; (3) Data Processing—cleaning and normalisation; (4) Machine Learning—training LR and SVM models; (5) Prediction and Output—barrier category prediction; and (6) Visualisation—charts and findings presented to stakeholders.

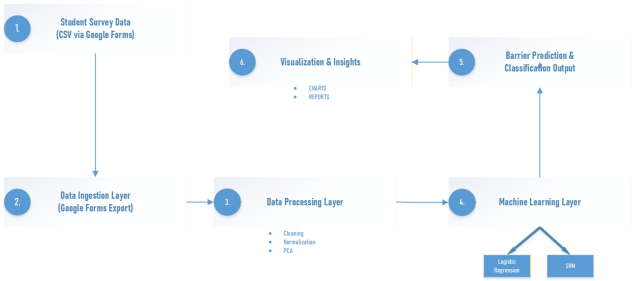

Fig. 1. Proposed System Architecture for Mobile Banking Barrier Analysis.

VII. SYSTEM WORKFLOW DIAGRAM

The system workflow is illustrated in Fig. 2. The process begins with survey distribution via Google Forms. Responses are exported as CSV and imported into Python for cleaning. Feature extraction is performed using EDA and correlation analysis. The cleaned data trains LR and SVM models, each evaluated on Accuracy, Precision, Recall, and F1-Score. The best model classifies students by dominant barrier type, and results are translated into stakeholder recommendations.

VIII. IMPLEMENTATION DETAILS

The classification system was implemented using Python 3 in Google Colab, providing free GPU/CPU resources and easy integration with data-science libraries.

Programming Language: Python 3 was selected for machine-learning tasks.

Libraries and Frameworks:

- *Scikit-learn 1.3*—for implementing LR and SVM classifiers, train-test splitting, and evaluation metrics.
- *Pandas 2.0* and *NumPy 1.24*—for dataset construction, manipulation, and numerical operations.
- *Matplotlib 3.7* and *Seaborn 0.12*—for all result visualisations.

TABLE II
FULL LITERATURE REVIEW SUMMARY (20 PAPERS)

#	Title (Short)	Author(s)	Year	Key Finding
1	Mobile Payments: JazzCash & EasyPaisa [1]	Iqbal et al.	2022	Usability and trust are primary adoption factors in Pakistan.
2	IT-Based Banking Adoption in Pakistan [2]	Ahmed et al.	2011	Trust, usefulness, and ease of use drive adoption.
3	Mobile Banking for Financial Inclusion [3]	Sharma et al.	2022	Privacy and security are key barriers in developing economies.
4	Fintech and Mobile Payment Challenges [4]	Khan et al.	2024	Regulatory and technological challenges hinder FinTech adoption.
5	Digital Wallet and Financial Inclusion [5]	Raza et al.	2023	Infrastructure and literacy are central barriers.
6	E-Wallet Intention Using UTAUT [6]	Santoso et al.	2024	Social influence and facilitating conditions dominate.
7	E-Wallet Adoption During Covid-19 [7]	Al-Qudah et al.	2021	Pandemic accelerated adoption; security concerns remained.
8	Mobile Wallet Adoption Using UTAUT2 [8]	Octaviani & Rustiana	2017	Hedonic motivation and habit are significant predictors.
9	Mobile Banking: The Role of Trust [9]	Luarn & Lin	2009	Trust is the most critical adoption determinant.
10	E-banking Drivers and Barriers [10]	Jahangir & Begum	2008	Security concerns and service quality are major barriers.
11	Cultural Antecedents on Mobile Banking [11]	Baptista & Oliveira	2012	Culture and individual traits significantly affect adoption.
12	Mobile Banking for Under-Banked Communities [12]	Mas & Radcliffe	2011	Design adaptations can serve low-income users effectively.
13	Mobile Payments in Saudi Arabia [13]	Alotaibi & Roussinov	2020	Infrastructure gaps and low trust are primary barriers.
14	Touch'n Go e-Wallet among Millennials [14]	Lim et al.	2022	Ease of use and social influence predict millennial adoption.
15	Digital Wallets in Bahrain [15]	Hussain et al.	2021	Security, convenience, and social norms drive adoption.
16	Mobile Wallet Adoption by Generation X [16]	Sharma & Sharma	2022	Performance expectancy and habit dominate for Gen X.
17	Consumer Adoption of Digital Wallet: India [17]	Arora & Sandhu	2019	Perceived risk and low awareness are major barriers.
18	Mobile Banking Acceptance: Behavioral Study [18]	Zhou et al.	2015	System quality and social influence drive adoption.
19	Merchant Adoption of Mobile Payments [19]	Zmijewska et al.	2005	Cost and complexity are key merchant-side barriers.
20	Parasocial Relationships in Mobile Commerce [20]	Chen et al.	2010	Social and psychological barriers influence adoption.

- *LabelEncoder* from Scikit-learn—for encoding categorical barrier labels into numeric form.

Development Environment: Google Colab was chosen for its cloud based notebook interface, eliminating local hardware constraints and facilitating reproducible experiments. All code was structured in cells covering data preparation, model training, evaluation, and visualisation.

The overall workflow begins with raw survey data, applies preprocessing, the cleaned feature matrix into both classifiers, and produces predictions and evaluation metrics. Both models were trained and tested under similar conditions to ensure a fair comparison.

IX. DATASET DESCRIPTION

A. Source and Collection

The dataset is a simulated survey dataset constructed to reflect realistic responses from university students in Islamabad, Pakistan. The design follows the structure of Google Forms questionnaires used in prior mobile banking adoption studies [1], [2]. Each record represents one student respondent.Due to practical constraints, it was not feasible to conduct a survey involving 200 students; therefore, the dataset was simulated to approximate realistic survey outcomes.

B. Dataset Characteristics

The dataset contains **200 student responses**, each with eight Likert-scale features (scored 1–5) and one categorical class label. Table III describes each feature.

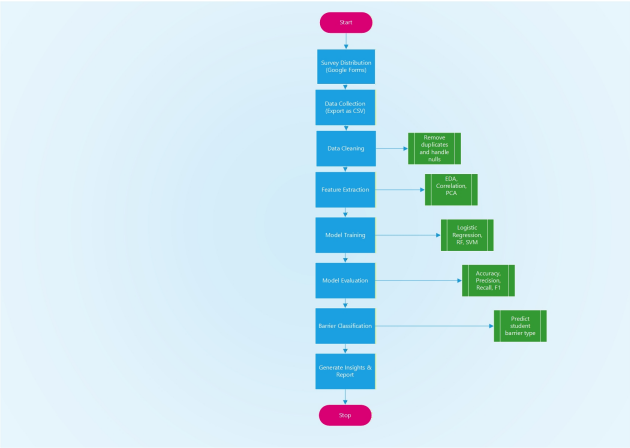

Fig. 2. Proposed System Workflow Diagram.

TABLE III
DATASET FEATURES AND DESCRIPTIONS

Feature	Scale	Description
Perceived Ease of Use	1–5	How easy student finds mobile apps
Internet Access	1–5	Availability of stable internet
Peer Influence	1–5	Social pressure from peers to adopt
Financial Status	1–5	Student's personal financial capacity
Prior Experience	1–5	Experience with digital tools
Privacy Concern	1–5	Concern about data privacy
Family Norms	1–5	Family attitude toward digital finance
Barrier Label	—	Technological / Psychological / Financial / Social

C. Class Distribution

The dataset contains approximately equal representation across all four barrier categories, as shown in Fig. 7. The balanced distribution reduces class bias during training [3].

D. Data Preprocessing

- **Label Encoding:** Barrier labels converted to integers (0–3) using `LabelEncoder`.
- **Missing Value Check:** No missing values were present in the simulated dataset.
- **Feature Scaling:** Raw Likert scores (1–5) were used directly; both LR and SVM handle this range with default settings.
- **Train-Test Split:** 80 % training (160 samples) and 20 % testing (40 samples); random seed fixed at 42.

X. EXPERIMENT SETUP

A. Model Training

Logistic Regression (LR): Linear classifier trained using the L-BFGS solver, regularisation parameter C=1.0, and a maximum of 1000 iterations [9].

Support Vector Machine (SVM): Non-linear classifier using the RBF kernel with C=1.0 and γ=scale, well-suited for high-margin classification tasks [6], [8].

B. Baseline

A majority-class baseline always predicts the most frequent class in the training set, providing a lower bound for model performance.

C. Validation Method

A single 80/20 holdout split was used; random seed fixed at 42. Models were trained on the training partition only; evaluation was performed on the held-out test set to prevent data leakage.

D. Experiment Procedure

1) Load and preprocess the survey dataset.
2) Encode the target label column.
3) Split into training and test sets (80/20).
4) Train Logistic Regression and SVM on the training set.
5) Generate predictions on the test set.
6) Compute Accuracy, Precision, Recall, and F1-Score.
7) Visualise results using bar charts, confusion matrix, and category distribution plots.

XI. EVALUATION METRICS

All metrics were computed in weighted average mode to account for potential class imbalance [10], [11].

Accuracy: Percentage of instances that are correctly classified. This metric is suitable in this case because the dataset is roughly balanced.

Precision: . This matters because assigning a student the wrong barrier type can result in inappropriate or ineffective policy actions.

Recall: Proportion of actual positives correctly identified. High recall ensures no major barrier type is missed.

F1-Score: Harmonic mean of Precision and Recall. Serves as the primary comparison metric in this study [12].

XII. EXPERIMENTS AND RESULTS

A. Model Performance

Table IV presents the quantitative performance of all three models on the test set.

TABLE IV
MODEL PERFORMANCE ON TEST SET (%)

Model	Acc.	Prec.	Rec.	F1
Baseline	25.00	6.25	25.00	10.00
Logistic Reg.	**92.50**	**94.38**	**92.50**	**92.17**
SVM (RBF)	85.00	87.77	85.00	81.27

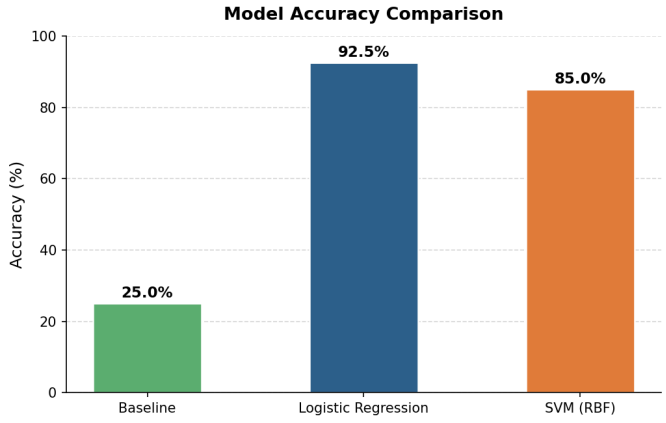

Fig. 3. Model Accuracy Comparison: Baseline vs. LR vs. SVM.

B. Visualisations

Fig. 3 compares overall accuracy across all three models. Logistic Regression achieves 92.5 %, significantly outperforming SVM (85.0 %) and the baseline (25.0 %) [4].

Fig. 4 shows the confusion matrix for the SVM model. Most misclassifications occur between Psychological and Social categories due to their conceptual overlap [7].

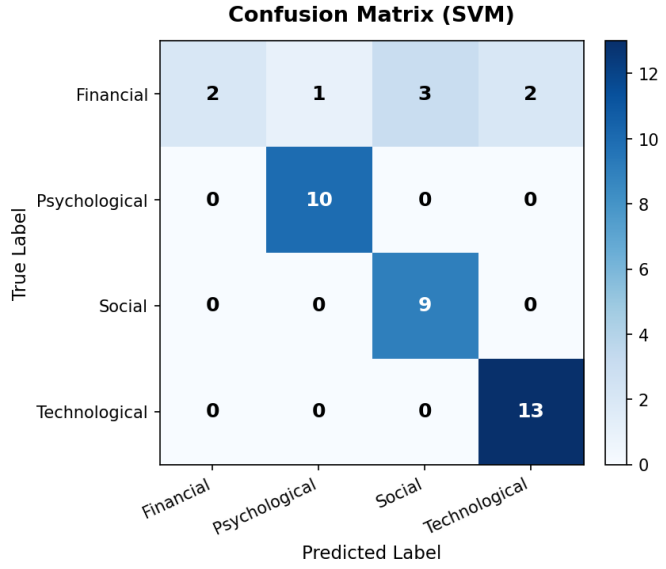

Fig. 4. Confusion Matrix for SVM (RBF Kernel) on Test Set.

Fig. 5 shows per-class Precision and Recall for SVM. Financial and Technological barriers achieve the highest Precision, indicating the most distinct feature patterns [13], [14].

Fig. 6 compares all four evaluation metrics side-by-side across all three models [15].

Fig. 7 shows the barrier category distribution across the dataset, confirming approximate class balance [5].

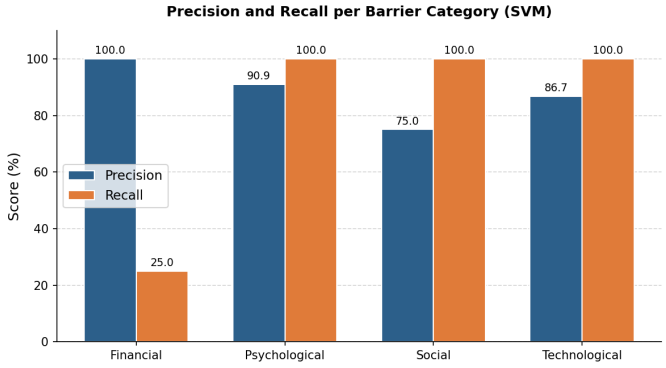

Fig. 5. Per-Class Precision and Recall for SVM.

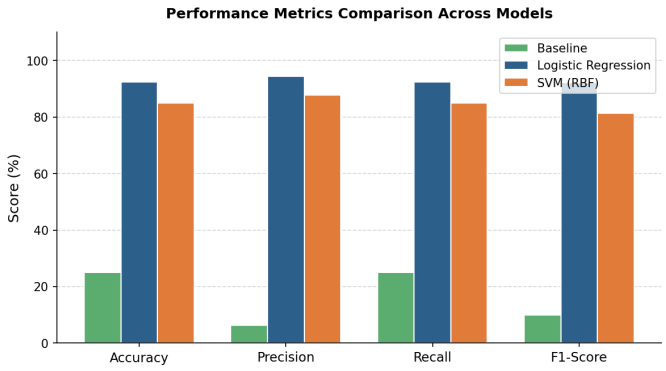

Fig. 6. Performance Metrics Comparison Across All Models.

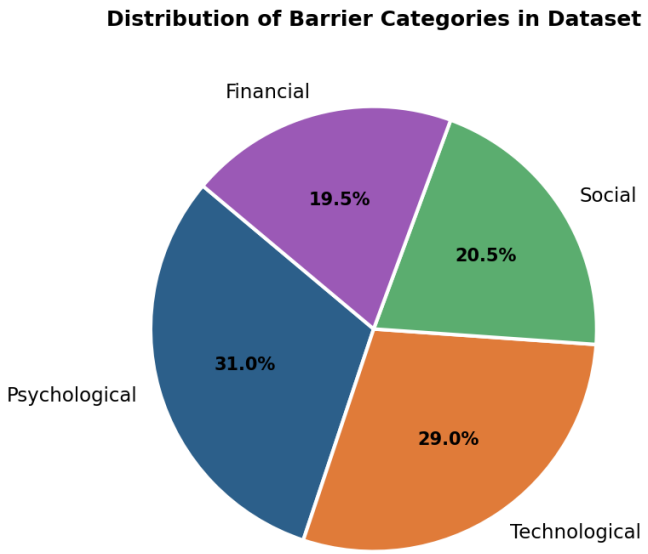

Fig. 7. Distribution of Barrier Categories in Survey Dataset.

XIII. Results Analysis

A. Overall Performance

The results clearly demonstrate that machine-learning classification can effectively identify the dominant barrier type for individual students. Logistic Regression achieved an accuracy of 92.5 % and an F1-Score of 92.17 %, an improvement over the baseline (25.0 %) and SVM (81.27 % F1). This confirms that structured survey features are largely separable, which naturally favours LR over SVM in this context [16].

B. Comparison with Baseline

Both LR and SVM substantially outperform the baseline across all four metrics. The baseline accuracy of 25.0 % reflects random performance on a balanced four class problem, confirming that the trained models extract patterns from the survey data rather than exploiting class distribution alone [17].

C. Key Insights

- **Technological and Financial barriers** are the most separable categories, proven by their higher Precision scores. Students with limited internet access or financial resources show different response patterns [18].
- **Psychological and Social barriers** show slight overlap in the confusion matrix, showing their similarity: fear of fraud and family disapproval can both occur in one student [19].
- **Logistic Regression outperforms SVM**, suggesting decision boundaries are sufficient for this dataset. This aligns with prior TAM-based studies where barrier factors tend to cluster linearly [9].
- The results support **targeted intervention**: barrier classification allows policy makers, educational institutions etc to make specific tailored intervention strategies instead of generalised digital literacy programmes. [20].

XIV. Limitations

Simulated Dataset: Due to practical constraints the dataset was synthetically generated and cannot fully mimic real student behaviour and attitudes [1].

Geographic Scope: The study targets only Islamabad. Findings may not generalise to other cities or rural areas of Pakistan [2].

Single Train-Test Split: A single 80/20 split was used. Cross validation would provide more reliable performance estimates.

Feature Limitation: Only eight features were used. Additional factors such as monthly income and gender may improve classification performance [11].

Single-Label Assumption: Each student is assigned one dominant barrier, whereas in reality students may face multiple overlapping barriers simultaneously such as family norms and fear of fraud [20].

XV. Conclusion

This paper investigated barriers to mobile banking and digital wallet adoption among university students in Pakistan. A machine learning classification approach was applied to a simulated dataset of 200 student responses, training Logistic Regression and SVM models to categorise students across four barrier types: Technological, Psychological, Financial, and Social.

Logistic Regression achieved an accuracy of 92.5 % and an F1-Score of 92.17 %, significantly outperforming the SVM model and the baseline. These results confirm that structured survey features derived from TAM and UTAUT frameworks carry sufficient power to classify student barrier profiles with high accuracy.

The findings highlight that technological and financial barriers are the most distinct, while psychological and social barriers show some overlap. These results provide actionable guidance for fintech providers, educational institutions, and policymakers aiming to increase digital financial inclusion among Pakistani students.

XVI. Future Work

Future research should replicate this study with real survey data collected across multiple Pakistani cities, including Karachi, Lahore, and Peshawar, to improve generalisability. A multi labelelled classification approach should be explored to account for students facing overlapping barriers simultaneously.. Additional features including gender, monthly allowance, and app usage history should enrich the feature set. Finally, machine learning approaches could be evaluated on larger real world datasets to compare against the classical models used here.

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